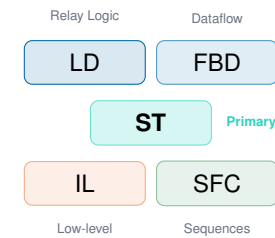


IEC 61131-3 Languages



WHAT THIS COURSE TEACHES

IEC 61131-3 is the international standard for PLC programming languages. Five languages, one standard: Ladder Diagram (LD), Function Block Diagram (FBD), Structured Text (ST), Instruction List (IL), and Sequential Function Chart (SFC). Every major PLC platform — Siemens, Allen-Bradley, Schneider, Beckhoff, Codesys-based vendors — implements some or all of these. An engineer who understands IEC 61131-3 can read control logic on any of them. An engineer who only knows one vendor's dialect cannot.

This course teaches all five languages with emphasis on Structured Text (the future of PLC programming) and Function Blocks (the reuse pattern every modern PLC project depends on). Lab exercises use the open Codesys runtime — no license, no hardware required.

CHAPTER SEQUENCE

1. **Standard Overview** — History of PLC programming before standardization, what IEC 61131-3 solved, vendor compliance levels.
2. **Program Organization Units** — Programs, Function Blocks, Functions, Tasks — the execution model and memory scoping rules.
3. **Data Types** — Elementary types, derived types, arrays, structures, enumerations, type safety rules.
4. **Ladder Diagram (LD)** — Contact/coil model, rung logic, timer and counter function blocks, safety circuit patterns.
5. **Function Block Diagram (FBD)** — Graphical dataflow, standard library blocks, PID function block wiring.
6. **Structured Text (ST)** — Syntax, control flow, assignment, iteration — full programming language for PLC logic.
7. **Sequential Function Chart (SFC)** — Step/transition model, action qualifiers, sequence design for batch processes.
8. **Standard Library** — IEC standard function blocks: TON, TOF, TP, CTU, CTD, MOVE, LIMIT, PID — correct usage and common mistakes.
9. **Safety & Best Practices** — PLCopen Motion guidelines, naming conventions, documentation requirements, code review.
10. **Code Lab** — Hands-on exercises: implement a motor interlock, a batch sequence, and a PID loop in Structured Text.

THE STORY

PLC programming has been balkanized for forty years. Each vendor had their own IDE, their own instruction set, their own way of doing timers. Engineers learned one platform and were expensive to retrain. IEC 61131-3 exists to end that — one standard, portable logic, transferable skills.

The momentum has shifted to Structured Text. New Codesys-based PLCs, Beckhoff TwinCAT, and even Siemens SCL all use ST as the primary language for complex logic. An engineer who only knows Ladder Diagram is increasingly working in a language that the industry is moving away from. This course brings engineers current — not by discarding LD, which is still the right tool

Certification Relevance

- **PLCopen Certified** — PLCopen offers function block and motion control certifications aligned directly to IEC 61131-3.
- **ISA CCST** — PLC programming languages and standard function blocks are tested in the control systems domain.
- **Siemens / Beckhoff Vendor Certs** — All major vendor certifications assume IEC 61131-3 as the foundation.

Next Steps (Certs)

1. Complete IEC 61131-3 + PID courses as a pair.
2. Install Codesys runtime (free) and complete all lab exercises.
3. Review PLCopen Function Block standard documents (free PDF at plcopen.org).
4. Register at plcopen.org for Function Block Certification exam.

10 Chapters 600+ Questions 5 Languages